

Over-Refusal of Benign Slovene Prompts in Gemma-3-12B Is a Criterion Shift

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1 Headline

On the reserved HARD evaluation set (1,093 safe and 349 harmful items, local Qwen3-14B judge), unedited Gemma-3-12B-IT applies a stricter refusal criterion in Slovene than in English on benign items: the difference-in-differences in criterion c is $+0.52$ $[0.38, 0.68]$, with no corresponding change in sensitivity (d' DiD = -0.003 $[-0.33, 0.29]$). False-alarm rates: Gemma EN 0.364 $\rightarrow$ SL 0.569 vs GaMS3 0.273. Source: Experiment 5 (Iteration 2). Evidence grade: **ESTIMATE**—the sign reverses under partial-as-refusal coding, and the readout gate has never passed. Outcome: not testable.

A secondary analysis finds an output-language effect (C-OUT): edited Gemma refuses $+2.43$ $[+0.89, +5.38]$ log-odds more when told to answer in Slovene than in English (PPI-corrected, Experiment 16, Iteration 5). C-OUT is also rated **ESTIMATE** (harmful-content adjudicator kappa = 0.12, anchor WEAK). These measurements indicate that multilingual safety evaluations should report false-alarm rates on benign items alongside attack-success rates.

2 Goal

The run tested whether Slovene-taught refusal in GaMS3-12B-Instruct transfers to English, whether English-only ablation (Heretic) removes Slovene refusal as effectively as English refusal, and what mechanism explains any residual Slovene refusal. GaMS3 and Gemma-3-12B-IT share a pretrained ancestor but diverge in safety-supervision language, forming a natural sibling pair.

3 What Was Tried

Iteration 1. Screen of four pre-registered claims on SCORE items: refusal DiD, dose contrast, base-checkpoint geometry, persona gating, Heretic transfer curve and prefill test. Killed alternates 2 and 3; dose contrast underpowered.

Iteration 2. Reserved FINAL evaluation on 1,300 pairs; signal-detection analysis on the HARD set (criterion shift found); teacher-inheritance screen (wording inherits, decisions do not); Heretic re-test on 200 item-matched probes; reconciliation of all iteration-1 screens.

Iteration 3. Screen repair of experiment 8 with fresh judges; fresh 300-item probe with DeBERTa judge (experiment 9); mechanism test of geometry and induction hypotheses (experiment 10); FINAL confirmation on reserved split with Hungarian arm (experiment 11); utility control on six tasks (experiment 12).

Iteration 4. Paid-judge readout completion (evaluation 3) pooling across item bodies; $2 \times 3 \times 3$ input $\times$ output language mechanism test (experiment 14); fine-grained lambda ladder on fresh 300-item body (experiment 15); held-out confirmation slot failed (GPU lost).

Iteration 5. C-OUT test on fresh 294-item body with PPI (experiment 16); RefusEU FINAL on 1,300 prompts per cell (experiment 17); R-INCAP safety evaluation with StrongREJECT (evaluation 4); paper-number audit (evaluation 5); novelty and fact check (research 2).

4 Key Results

Finding	Source	Grade	Number
Criterion shift (originals, HARD)	Exp 5, Iter 2	ESTIMATE	c DiD = +0.52 [0.38, 0.68]
C-OUT output-language effect	Exp 16, Iter 5	ESTIMATE	OUT _{Gemma} = +2.43 [0.89, 5.38]
Cross-model C-OUT contrast	Exp 16, Iter 5	ESTIMATE	dOUT = −2.31 [−5.34, −0.68]
R-BASE (lag is baseline)	Exp 15, Iter 4	Supported	$G_{3,\text{edit}}$ = +0.47 [−0.96, 2.14]
R-INCAP (reserve = safety)	Eval 4, Iter 5	ESTIMATE	OUT _U = 2.61 [2.20, 3.07]
PF index (deflection share)	Eval 4, Iter 5	ESTIMATE	Δ PF = 0.46 [0.23, 0.65]
Utility: no SL damage	Exp 12, Iter 3	OK	Asymmetry $A > 0$ all cells
RefusEU FINAL edit effect	Exp 17, Iter 5	ROBUST	ASR Δ 0.657–0.937
Identity calibration	Exp 5, Iter 2	PASS	DiD _{id} = 3.46 [2.55, 4.62]

The count of 8 of 9 compliance-valid rows with a criterion CI below zero was a post-hoc refinement. Judge-agreement kappa ranged from 0.50 to 1.0 per cell (Qwen3 vs gemini, DEV split, unedited rows).

5 What Failed

Dead alternates. Base-checkpoint geometry (alt 2): $\Delta d'$ sign depends on layer. Persona gating (alt 3): manipulation check fails, premise contradicted. Prefill-depth signature (alt 4): uncitable—no readout exceeds $\kappa = 0.60$ on prefill continuations.

Geometry and induction hypotheses. Nearly collinear at own- L^* ($\cos = 0.986\text{--}0.991$); layer-dependent at common layer. The Slovene-perpendicular component does not induce refusal.

Readout gate. Never passed in any iteration. Best judge κ on edited text: 0.43 (gemini). Specificity on edited Slovene rows: 0.42–0.73.

Experiment 13 (held-out confirmation). GPU lost 25 minutes into run; all generation and labelling cancelled.

Frozen-core Heretic (bf16, 80 trials, Heretic’s own selection). Never executed across five iterations. Natural RefusEU reserved-split under the edit: also never executed.

Open questions. All adjudication is author-LLM, not human. The frozen-core Heretic procedure and the natural reserved-split evaluation remain unexecuted. If R-BASE is correct, the lag is a property of model selection (GaMS3 was trained to refuse less in Slovene), not of the ablation procedure. The C-OUT finding, rated ESTIMATE, deserves pre-registered confirmation with a validated readout and human adjudication. The harmful-content adjudicator kappa (0.12 in experiment 16; 0.59 in evaluation 4) remains below the gate. The novelty verdict is PARTIALLY OCCUPIED after Addagada (arXiv 2609.08373) and Nguyen et al. (arXiv 2608.26186); the narrowed design (crossing both factors with safety endpoints on a sibling pair under partial-dose ablation) remains unoccupied (medium-low confidence). The full report is 31 pages.